

## EDUCATION & APPOINTMENTS

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- Postdoctoral Fellow, Miller Institute for Basic Research in Science, UC Berkeley 2025-present  
Advisors: Dr. Noah Whiteman, Dr. Benjamin Blackman
- Ph.D., Population Biology, University of California, Davis 2019-2025  
Dissertation: "On Wavelets and Walnuts: Evolutionary Genomics of Hybridization and Heterodichogamy." Advisor: Dr. Graham Coop
- Occupational Trainee, Ortiz-Barrientos Lab, University of Queensland 2018-2019
- B.Sc. (Honours) Plant Biology, University of British Columbia 2014-2018  
Honours thesis: "On the origin of orphan hybrids between *Aquilegia flavescens* and *A. formosa*"  
Honours thesis advisor: Dr. Quentin Cronk  
Undergraduate Researcher, Schluter Lab, 2016-2017

## PUBLICATIONS

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- Groh JS** (2026). Heterodichogamy. *Current Biology* (In Press, scheduled publication Nov 2<sup>nd</sup>, 2026).
- Groh JS**, Soares dos Santos MF, Avila de Dios E, Ackerman G\*, Solares E, Iturrieta RA, Focht E, Seymour D, Gaut BS, Arpaia ML, and Coop G (2026). Balanced polymorphism in a floral transcription factor underlies an ancient rhythm of daily sex alternation in avocado. *PNAS* 123: e26068761. <https://doi.org/10.1073/pnas.2606876123>.
- Roesti M, **Groh JS**, Jones F, Peichel CL, and Schluter D (2026). Reply to MacColl: Repeated evolution from standing genetic variation. *PNAS* 123: e2605003123. <https://doi.org/10.1073/pnas.2605003123>.
- Groh JS**, Ackerman G\*, Wenzel M, and Coop G (2025). Distinct haplotypes and reversed dominance at a single-gene balanced polymorphism controlling heterodichogamy in two genera of wingnuts. *Current Biology* 35: 5192-5209.e11. <https://doi.org/10.1016/j.cub.2025.09.061>.
- Roesti M, **Groh JS**, Jones F, Peichel CL, and Schluter D (2025). A species interaction kickstarts ecological speciation in allopatry. *PNAS* 122: e2506625122. <https://doi.org/10.1073/pnas.2506625122>.
- Groh JS**, Vik DC\*, Davis M, Stevens KA, Brown PJ, Langley CH, and Coop G (2025). Ancient structural variants control sex-specific flowering time morphs in walnuts and hickories. *Science* 387: eado5578. [Cover article]. DOI: 10.1126/science.ado5578.
- Langdon QK, **Groh JS**, Aguillon SM, Powell DL, Gunn T, Payne C, Baczenas JJ, Donny A, Dodge TO, Du K, Scharf M, Ríos-Cárdenas O, Gutiérrez-Rodríguez C, Morris M, and Schumer M (2024). Swordtail fish hybrids reveal that genome evolution is surprisingly predictable after initial hybridization. *PLoS Biology* 22: e3002742. <https://doi.org/10.1371/journal.pbio.3002742>.
- Groh JS** and Coop G (2023). The temporal and genomic scale of selection following hybridization. *PNAS* 121: e2309168121. <https://doi.org/10.1073/pnas.2309168121>.

James ME, Allsopp RN\*, **Groh JS**, Kaur A, Wilkinson MJ, and Ortiz-Barrientos D. (2023). Uncovering the genetic architecture of parallel evolution. *Molecular Ecology* 32: 5575-5589.  
<https://doi.org/10.48610/40dec7a>.

Roesti M, **Groh JS**, Blain SA, Huss M, Rassias P, Bolnick DI, Stuart YE, Peichel CL, and Schluter D (2022). Species divergence under competition and shared predation. *Ecology Letters* 26: 111-123. [Cover article].  
<https://doi.org/10.1111/ele.14138>.

James ME, Arenas-Castro H, **Groh JS**, Allen SL, Engelstädter J, and Ortiz-Barrientos D (2021). Highly replicated evolution of parapatric ecotypes. *Molecular Biology and Evolution* 38: 4805-4821.  
<https://doi.org/10.1093/molbev/msab207>.

**Groh JS** and Cronk QCB (2020). Phenotypic evidence for an extensive mosaic hybrid zone between two species of columbine, *Aquilegia flavescens* and *A. formosa*. *Botany* 98: 459-467.  
<https://doi.org/10.1139/cjb-2020-0015>.

**Groh JS**, Percy DM, Björk CR, and Cronk QCB (2019). On the origin of orphan hybrids between *Aquilegia formosa* and *Aquilegia flavescens*. *AoB Plants* 11: ply071. [Editor's Choice].  
<https://doi.org/10.1093/aobpla/ply071>.

\*Undergraduate mentee

## FELLOWSHIPS AND RESEARCH GRANTS

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|                                                                                  |            |
|----------------------------------------------------------------------------------|------------|
| Miller Postdoctoral Research Fellowship, UC Berkeley (\$325,000)                 | 2025-2028  |
| Rausser Research Seed Grant, UC Berkeley College of Natural Resources (\$10,000) | 2026       |
| Davis Botanical Society Research Grant (\$4,000)                                 | 2024, 2025 |
| National Science Foundation Graduate Research Fellowship (\$138,000)             | 2019-2024  |
| UC Natural Reserves Graduate Student Research Grant (\$2,500)                    | 2024       |
| American Society of Plant Taxonomists Research Grant (\$1,200)                   | 2024       |
| Center for Population Biology Research and Travel Award, UC Davis (\$1,500)      | 2024       |
| Dean's Distinguished Graduate Fellowship, UC Davis (\$27,000)                    | 2019       |
| American Australian Association Education Fellowship (\$30,000)                  | 2018       |
| Mackenzie King Open Scholarship (\$13,000, declined due to funding overlap)      | 2018       |
| International Undergraduate Summer Research Award, UBC (\$6,000)                 | 2017       |
| Botany and Zoology Student Research Award, UBC (\$1,000)                         | 2017       |
| Walter Lewis Research Award in Plant Biodiversity, UBC (\$1,000)                 | 2017       |
| National Science and Engineering Research Council SURE Award (\$4,500)           | 2016       |

## AWARDS & DISTINCTIONS

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|--------------------------------------------------------------------------------|------|
| James Crow Early Career Research Award (Finalist), Genetics Society of America | 2026 |
| Merton Love Award*, Department of Evolution & Ecology, UC Davis                | 2025 |
| Best Poster, Bay Area Population Genetics Conference, UC Berkeley              | 2024 |
| C.K. Choi Award and Wesbrook Scholar**, UBC                                    | 2017 |
| International Community Achievement Award, UBC                                 | 2015 |
| Outstanding International Student Award, UBC                                   | 2014 |
| International Major Entrance Scholarship, UBC                                  | 2014 |

\* "for Outstanding Dissertation in Ecology and Evolution"

\*\* "UBC's most prestigious designations offered to senior students"

## PRESENTATIONS

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### *Invited Talks*

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|----------------------------------------------------------------------------------------------------|------|
| California Academy of Sciences. San Francisco, CA USA                                              | 2026 |
| Institute of Ecology & Evolution, University of Bern. Bern, Switzerland                            | 2026 |
| Zoology and Evolutionary Biology Institute, University of Basel. Basel, Switzerland                | 2026 |
| TiBE-STRIVE: Evolutionary Role of Structural Genomic Variation. Vila do Conde, Portugal (*Plenary) | 2026 |
| James F. Crow Early Career Researcher Award Symposium, PEQG. Asilomar, CA USA                      | 2026 |
| Integrative Biology Department, UC Berkeley. Berkeley, CA USA                                      | 2026 |
| Evolution of Mating Systems guest lecture, University of Lille. (virtual)                          | 2025 |
| University and Jepson Herbaria, UC Berkeley. Berkeley, CA USA                                      | 2025 |
| Center for Theoretical and Evolutionary Genetics, UC Berkeley. Berkeley, CA USA                    | 2025 |
| Merton Love Award Seminar, UC Davis. Davis, CA USA                                                 | 2025 |
| Wright Lab, University of Toronto. Toronto, ON Canada                                              | 2024 |
| Schumer Lab, Stanford University. Stanford, CA USA                                                 | 2024 |

### *Contributed Talks*

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|---------------------------------------------------------------------------------|------|
| Department of Molecular and Cell Biology retreat, UC Berkeley. Asilomar, CA USA | 2025 |
| Evolution Meeting. Athens, GA USA                                               | 2025 |
| Virtual Evolution Meeting                                                       | 2025 |
| Sex Determination Symposium, CSEE. Vancouver, BC Canada                         | 2024 |
| The Allied Genetics Conference, Washington, DC USA                              | 2024 |
| Evolution Meeting, Albuquerque, NM USA                                          | 2023 |
| Bay Area Population Genetics Conference, Stanford University, Stanford, CA USA  | 2022 |
| Center for Population Biology, UC Davis, Davis, CA USA                          | 2021 |
| Virtual Evolution Meeting                                                       | 2020 |

### *Poster Presentations*

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|---------------------------------------------------------------------------------|------|
| Department of Plant and Microbial Biology retreat, UC Berkeley, Berkeley CA USA | 2026 |
| Department of Molecular and Cell Biology retreat, UC Berkeley, Berkeley CA USA  | 2026 |
| Bay Area Population Genetics Conference, Stanford University, Stanford, CA USA  | 2025 |
| Society for Molecular Biology and Evolution Conference, Puerto Vallarta, Mexico | 2024 |
| Population, Evolutionary and Quantitative Genetics Conference, Asilomar, CA USA | 2022 |
| Evolution Meeting, Providence, RI USA                                           | 2019 |

## TEACHING & MENTORSHIP

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|--------------------------------------------------------------------------|-----------|
| Teaching Assistant, EVE 102 Population & Quantitative Genetics, UC Davis | 2024      |
| Mentor, Ecology & Evolution Graduate School Preview Program, UC Davis    | 2020-2024 |
| Instructor of Record, EVE 100 Introduction to Evolution, UC Davis        | 2023      |
| Teaching Assistant, EVE 100 Introduction to Evolution, UC Davis          | 2021      |

### Undergraduate mentees

Felix Vik, enrolled Ph.D. program in Integrative Biology, Oregon State University  
Jennifer Lee, enrolled in Master's program in Marine Biology and Ecology, University of Miami  
Gracie Ackerman, Makayla Davis, Haley Hoefer, Aariya Shergill, Natalie Su, Kai Pessanha, Emma Konecna, Xinyu Song

## SERVICE, LEADERSHIP & OUTREACH

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|                                                                                         |              |
|-----------------------------------------------------------------------------------------|--------------|
| Miller Institute Symposium Planning Committee                                           | 2026         |
| Meet the Millers, for UC Berkeley undergraduates interested in STEM careers             | 2026         |
| Co-organizer, Evolutionary Genomics Reading Group, UC Berkeley                          | 2025-2026    |
| 'Ask a Geneticist' at 'Expect Greater' UC Davis fundraising event                       | 2024         |
| Lead organizer, Bay Area Population Genetics Conference, UC Davis                       | 2023         |
| DEIJ Leader Fellow, College of Biological Sciences, UC Davis                            | 2023         |
| Student Diversity Committee chair, Population Biology Graduate Group, UC Davis          | 2022-2023    |
| Student Representative Organizer, Population Biology Graduate Group, UC Davis           | 2022-2023    |
| Steering Committee student representative, Population Biology Graduate Group, UC Davis  | 2021-2022    |
| Graduate Student Conference lead organizer, Population Biology Graduate Group, UC Davis | 2019         |
| Scholarship application reviewer, American Australian Association                       | 2019-present |

I have peer-reviewed manuscripts for:

*Science, PNAS, Genetics, New Phytologist, Molecular Ecology, BMC Genomics, G3, Taxon, Ecology & Evolution*

## SELECTED MEDIA COVERAGE

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*Scientific American*. July 29, 2026. "Avocado flowers switch sex daily – now we know why."

<https://www.scientificamerican.com/article/avocado-flowers-switch-sex-daily-now-we-know-why/>.

Coverage of Groh et al. 2026, *PNAS*.

*The Wall Street Journal*. Aug 6, 2026. "Breakthrough on how avocado trees change sexes could cut years from breeding cycle." <https://www.wsj.com/science/biology/avocado-sex-change-study-6fdfab1d>. Coverage

of Groh et al. 2026, *PNAS*.

*Science*. In Other Journals. Sept 17, 2026. "Balancing Acts." Simonti C et al. DOI: 10.1126/science.aem4011.

Coverage of Groh et al. 2026, *PNAS*.

*SFGATE*. Jul 31, 2026. "Groundbreaking discovery could remake California's avocado breeding." Coverage of

Groh et al. 2026, *PNAS*.

*Current Biology*. Nov 3, 2025. "Evolution: Disassortative mating going nuts." Contreras-Garrido A and Castric

V. [https://www.cell.com/current-biology/fulltext/S0960-9822\(25\)01301-6](https://www.cell.com/current-biology/fulltext/S0960-9822(25)01301-6). Coverage of Groh et al.

2025, *Curr. Biol*.

*IFL Science*. Jan 7, 2025. "Walnut Trees' Flowers Have Been Changing Sex Every Season For 40 Million Years."

*IFL Science*. [https://www.iflscience.com/walnut-trees-flowers-have-been-changing-sex-every-season-for-40-](https://www.iflscience.com/walnut-trees-flowers-have-been-changing-sex-every-season-for-40-million-years-77513)

[million-years-77513](https://www.iflscience.com/walnut-trees-flowers-have-been-changing-sex-every-season-for-40-million-years-77513). Coverage of Groh et al. 2025, *Science*.

*KCRW* (NPR station in Los Angeles). Aug 5, 2026. Interviewed for Press Play with Madeleine Brand. Coverage of Groh et al. 2026, *PNAS*.